

Optimal Placement of Bearing-only Sensors for Target Localization

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Abstract—We investigate optimal placements of multiple bearing-only sensors for target localization in both 2D and 3D spaces. The target is assumed to be static, and sensor-target ranges are arbitrary but fixed. The Fisher information matrix is used to characterize the localization uncertainty. By employing frame theory, we show that there are two types of optimal sensor placements, regular and irregular. Necessary and sufficient conditions of optimal placements are presented. It is proved that an irregular optimal placement can be converted to a regular one in a lower dimensional space. We furthermore propose explicit algorithms to construct some important specific regular optimal placements.

I. INTRODUCTION

Optimal sensor placement is a fundamental problem for many applications such as cooperative control of sensor networks [1], cooperative target tracking [2], [3], [4] and path planning [5], [6]. The existing work on optimal sensor placement covers several sensor types including bearing-only [7], [8], range-only [4], [7], time-of-arrival-based [7] and signal strength based [9]. In this paper, we focus only on bearing-only sensors, which can measure only bearing information of a target. The reason why we are interested in bearing-only sensors is partly due to a particular type of sensor: camera. A monocular camera inherently is bearing-only because depth information of scenes are totally lost during perspective projection. Nowadays cameras play important roles in a wide range of applications such as vision-based control, navigation and guidance for robotics and unmanned aerial vehicles [10].

Recently, a comprehensive analysis of bearing-only sensor placement in 2D is presented in [7], [8]. The authors consider a static localization problem involving a single static target and multiple bearing-only sensors. The sensor-target ranges are assumed to be fixed but arbitrary. The target localization uncertainty is characterized by using Fisher information matrix (FIM). Necessary and sufficient conditions for achieving optimal placements are proposed. The authors also present methods to construct optimal placements with 2, 3 and 4 sensors. It is also shown how to construct optimal placements of arbitrary number of sensors with identical sensor-target ranges. However, it is not solved how to construct optimal placements of arbitrary number of sensors with arbitrary but fixed sensor-target ranges in 2D. Moreover, optimal

placement of bearing-only sensors in 3D space is still an open problem to our knowledge.

In this paper, we generalize the conclusions on bearing-only placements in [7], [8] to 3D cases. It is also assumed that the sensor-target ranges are arbitrary but fixed. We model the measurement of a bearing-only sensor as a unit-length vector, which enables us to formulate optimal bearing-only placement problem in 3D or even higher dimensions. The optimality criterion is chosen as minimizing the Frobenius norm of the FIM. We show that the optimality analysis heavily relies on *frame theory* [11], [12], [13], [14], [15].

The contributions of this paper are as the follows. Frame theory is introduced and applied to optimal placement problem of bearing-only sensors in both 2D and 3D. It is shown that there are two types of optimal placements: regular and irregular. We propose necessary and sufficient conditions of optimal placements. We show that an irregular optimal placement problem can be converted to a regular one in a lower dimensional space. Explicit constructions of several important specific placements are also presented.

II. PRELIMINARIES TO FRAME THEORY

Frames can be defined in any Hilbert spaces. We focus only on Euclidean space $\mathbb{R}^d$. A sequence $\{\varphi_i\}_{i=1}^N \subset \mathbb{R}^d$ with $N \geq d$ is called a *frame* if there exist constants $0 < \alpha \leq \beta < +\infty$ such that

$$\alpha \|v\|^2 \leq \sum_{i=1}^N |\langle v, \varphi_i \rangle|^2 \leq \beta \|v\|^2 \quad (1)$$

for all $v \in \mathbb{R}^d$, where $\langle *, * \rangle$ denotes the inner product between two vectors, and $\|*\|$ the Euclidean norm. In $\mathbb{R}^d$, we have $\|v\|^2 = v^T v$ and $|\langle v, \varphi_i \rangle|^2 = (v^T \varphi_i)^2 = v^T \varphi_i \varphi_i^T v$. α and β are called the *frame bounds*. A frame is *tight* when $\alpha = \beta$. From (1) the frame $\{\varphi_i\}_{i=1}^N$ is tight if and only if $v^T \left(\sum_{i=1}^N \varphi_i \varphi_i^T \right) v = \alpha v^T v$ for all $v \in \mathbb{R}^d$, which is equivalent to

$$\sum_{i=1}^N \varphi_i \varphi_i^T = \alpha I, \quad (2)$$

where I is the identity matrix with proper dimension. Taking trace on both sides of (2) yields $\alpha = \frac{1}{d} \sum_{i=1}^N \|\varphi_i\|^2$. One research topic in frame theory is to find a tight frame $\{\varphi_i\}_{i=1}^N$ solving (2) with prescribed norms. The following lemma [12] gives the existence condition on solutions to (2).

Lemma 1 (Fundamental Inequality): In $\mathbb{R}^d$, given a positive sequence $\{a_i\}_{i=1}^N$, there exists a tight frame $\{\varphi_i\}_{i=1}^N \subset$

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$\mathbb{R}^d$ solving (2) with $\|\varphi_i\| = a_i$ for all $i \in \{1, \dots, N\}$ if and only if

$$\max_{i=1, \dots, N} a_i^2 \leq \frac{1}{d} \sum_{i=1}^N a_i^2. \quad (3)$$

The inequality (3) is called as *fundamental inequality* of tight frames [12]. The sequence $\{a_i\}_{i=1}^N$ is called *regular* in $\mathbb{R}^d$ if (3) is satisfied, otherwise, is called *irregular*. The irregularity of an irregular sequence is defined as below [12].

Lemma 2 (Irregularity): For any irregular sequence $\{a_i\}_{i=1}^N$ with $a_1 \geq \dots \geq a_N > 0$, and any integer d with $1 \leq d \leq N$, there always exists a unique index k_0 with $1 \leq k_0 \leq d-1$ such that

$$a_{k_0}^2 > \frac{1}{d - k_0 + 1} \sum_{i=k_0}^N a_i^2, \\ a_{k_0+1}^2 \leq \frac{1}{d - k_0} \sum_{i=k_0+1}^N a_i^2.$$

Then k_0 is called as the *irregularity* of $\{a_i\}_{i=1}^N$ with respect to d . The irregularity of a regular sequence is defined as zero.

It is notable that the irregularity of a sequence is evaluated with respect to a specific integer d . From Lemma 2, the irregularity of $\{a_i\}_{i=k_0+1}^N$ with respect to $d - k_0$ is zero. That means a subset of an irregular sequence can be a regular one with respect to a smaller integer.

Constructing a tight frame solving (2) is recognized as notoriously difficult. By defining *frame force* and *frame potential* [11], [12], solving (2) can be reformulated as an optimization problem. The frame force $f(\varphi_i, \varphi_j) \in \mathbb{R}^d$ applied to φ_i by φ_j is defined as [12]

$$f(\varphi_i, \varphi_j) = 2\varphi_i^T \varphi_j (\varphi_i - \varphi_j).$$

Then the total frame potential of $\{\varphi_i\}_{i=1}^N$ is given by [12]

$$P(\{\varphi_i\}_{i=1}^N) = \sum_{i=1}^N \sum_{j=1}^N (\varphi_i^T \varphi_j)^2. \quad (4)$$

Lemma 3: In $\mathbb{R}^d$, given a frame $\{\varphi_i\}_{i=1}^N$ with $\|\varphi_i\| = a_i > 0$. Then the frame potential is bounded below by

$$P(\{\varphi_i\}_{i=1}^N) \geq \frac{1}{N} \left(\sum_{i=1}^N a_i^2 \right)^2.$$

Furthermore, the lower bound is achieved if and only if $\{\varphi_i\}_{i=1}^N$ is tight.

Lemma 3 enables us to find a tight frame by minimizing the frame potential. When the norms $\{a_i\}_{i=1}^N$ is irregular, no tight frames exist and the lower bound of the frame potential is unachievable. However, we can still find the *tightest* (but not tight) frame, a minimizer of the frame potential, by using the following lemma [12].

Lemma 4: In $\mathbb{R}^d$, let $\{a_i\}_{i=1}^N$ be a positive decreasing sequence of irregularity k_0 . Any minimizer of the frame potential is of the form

$$\{\varphi_i\}_{i=1}^N = \{\varphi_i\}_{i=1}^{k_0} \cup \{\varphi_i\}_{i=k_0+1}^N,$$

where $\{\varphi_i\}_{i=1}^{k_0}$ is an orthogonal set with φ_i 's mutually orthogonal; $\{\varphi_i\}_{i=k_0+1}^N$ forms a tight frame in the orthogonal complement of $\text{span}(\{\varphi_i\}_{i=1}^{k_0})$. Any local minimizer of the frame potential is also a global minimizer.

III. PROBLEM FORMULATION

Consider a static target and multiple bearing-only sensors in $\mathbb{R}^d$ with $d = 2$ or 3 . The sensor-target ranges are fixed but arbitrary, i.e., sensors are restricted on concentric spheres centered at the target. Denote the position of the target and the sensors respectively as $p \in \mathbb{R}^d$ and $s_i \in \mathbb{R}^d$ with $i \in \{1, \dots, N\}$. Given an initial estimation of the target position, our task is to determine sensors' position such that the relative sensor-target placement can minimize the target localization uncertainty.

A. Bearing-Only Measurement Model

By convention, the bearing-only measurement is modeled as an angle (azimuth) in 2D and two angles (azimuth and altitude) in 3D. The drawback of this kind of model is the model complexity increases dramatically as dimension increases. Motivated by this, we model the measurement of the bearing-only sensor i as a unit-length vector corrupted by additive noises:

$$\hat{m}_i(s_i, p) = m_i(s_i, p) + w_i,$$

where

$$m_i(s_i, p) = \frac{s_i - p}{\|s_i - p\|}.$$

The measurement error $w_i \in \mathbb{R}^d$ is assumed to be zero-mean Gaussian distributed with covariance matrix as $\Sigma_i = \sigma_i^2 I$.

B. Fisher Information Matrix

The target position p is the unknown but deterministic parameters to be estimated. Given measurements $m_i(s_i, p)$ of p , the FIM can describe the information of p carried by $m_i(s_i, p)$. Under the assumption that the noises of different sensor measurements are uncorrelated, the FIM can be computed by

$$F = \sum_{i=1}^N (\nabla_p m_i)^T \Sigma_i^{-1} \nabla_p m_i, \quad (5)$$

where $\nabla_p m_i$ is the Jacobian of $m_i(s_i, p)$ with respect to p :

$$\nabla_p m_i = -\frac{1}{\|r_i\|} (I - m_i m_i^T)$$

with $r_i = s_i - p$. Substituting the above equation into (5) yields

$$F = \sum_{i=1}^N c_i^2 (I - m_i m_i^T) \quad (6)$$

with $c_i = \frac{1}{\sigma_i \|r_i\|} > 0$.

It is notable that c_i 's are scale factors of F . It will be shown that the relative values between c_i 's are critical for the optimal placement, the absolute values, however, are not. The

quantity c_i consists of σ_i and $\|r_i\|$. If $\sigma_i = \sigma_j$ for all $i \neq j$, σ_i can be removed from the formula of F without influencing the final optimal placement. But to make our analysis more general, we will not assume σ_i 's to be identical.

To end this subsection, we would like to point out that the FIM derived in [7], [8] for 2D cases is a special version of the FIM in (6). To see this, express the unit-length vector m_i as $m_i = (\sin \phi_i, \cos \phi_i)^T$ with ϕ_i as the azimuth angle in 2D. Then (6) becomes

$$F = \sum_{i=1}^N \frac{1}{\sigma_i^2 \|r_i\|^2} \begin{pmatrix} 1 - \sin^2 \phi_i & -\sin \phi_i \cos \phi_i \\ -\sin \phi_i \cos \phi_i & 1 - \cos^2 \phi_i \end{pmatrix},$$

which is the same as the FIM shown in [7], [8].

C. Optimality Criterion

The Cramer-Rao Lower Bound (CRLB) states that the variance of any unbiased estimators is bounded below by the inverse of FIM. In order to minimize the target localization uncertainty, a widely adopted optimality criterion [4], [5], [7], [8] is to maximize the determinant $\det F$, which can be interpreted as minimizing the CRLB or the volume of the uncertainty ellipsoid given by $x^T F^{-1} x = 1$. From information theory perspective, maximizing $\det F$ can be interpreted as maximizing the information amount of the target carried by bearing-only measurements. For 2D optimal placements, the closed-form expression of $\det F$ can be computed and used for maximization. But for 3D cases the formula of $\det F$ is mathematically intractable. Motivated by this, we present a different reasonable optimality criterion.

Let $\{\lambda_j\}_{j=1}^d$ be the eigenvalues of F . Since $\{c_i\}_{i=1}^N$ is fixed, $\sum_{j=1}^d \lambda_j = \text{tr} F = (d-1) \sum_{i=1}^N c_i^2$ is constant. Thus it is straightforward to see

$$\det F = \prod_{j=1}^d \lambda_j \leq \bar{\lambda}^d,$$

where $\bar{\lambda} = \frac{1}{d} \sum_{j=1}^d \lambda_j = \frac{d-1}{d} \sum_{i=1}^N c_i^2$. The determinant $\det F$ achieves the maximum if and only if $\lambda_j = \bar{\lambda}$ for all $j \in \{1, \dots, d\}$, which is equivalent to

$$F - \bar{\lambda} I = 0.$$

We naturally define $\|F - \bar{\lambda} I\|$ as the distance from optimality of a placement. $\|\cdot\|$ denotes the Frobenius norm of a matrix. Generally speaking, when the distance from optimality is minimized, optimal placements are achieved. The optimal placement problem is reformulated as

$$\{m_i^*\}_{i=1}^N = \arg \min_{\{m_i\}_{i=1}^N} \|F - \bar{\lambda} I\|^2. \quad (7)$$

Furthermore, since $\|F - \bar{\lambda} I\|^2 = \text{tr}(F - \bar{\lambda} I)^2 = \|F\|^2 - d\bar{\lambda}^2$, (7) is equivalent to

$$\{m_i^*\}_{i=1}^N = \arg \min_{\{m_i\}_{i=1}^N} \|F\|^2. \quad (8)$$

It is notable that maximizing $\det F$ is rigorously equivalent to minimizing $\|F\|^2$ in 2D cases because $\det F =$

$\frac{1}{2} [(\text{tr} F)^2 - \text{tr}(F^2)] = \frac{1}{2} [(\text{tr} F)^2 - \|F\|^2]$ in 2D. However, this rigorous equivalence does not hold in 3D cases in general.

D. Properties of the FIM

Before solving the problem (8), we first analyze certain properties of F .

Proposition 1 (Singular Placement): F is positive (semi-) definite. F is singular if and only if $\{m_i\}_{i=1}^N$ are collinear.

Proof: Let $x \in \mathbb{R}^d$ be an arbitrary unit-length vector. Because $|x^T m_i| \leq 1$, we have

$$x^T F x = \sum_{i=1}^N c_i^2 [1 - (x^T m_i)^2] \geq 0.$$

Clearly $x^T F x = 0$ if and only if $|x^T m_i| = 1$ for all $i \in \{1, \dots, N\}$. Such x exists if and only if m_i 's are collinear. Thus F is singular if and only if m_i 's are collinear. The rank of a singular F is $d - 1$. ■

Geometrically, F is singular when all sensors and the target are located on the same line, along which the localization uncertainty is infinite. This kind of singular placement is the worst and should be avoided.

Proposition 2 (Eigenvalue Invariance): F is invariant to the sign change of m_i for all $i \in \{1, \dots, N\}$. The eigenvalues of F are invariant to any orthogonal transformations over $\{m_i\}_{i=1}^N$.

Proof: Since $m_i m_i^T = (-m_i)(-m_i)^T$ for all $i \in \{1, \dots, N\}$, $F = \sum_{i=1}^N c_i^2 (I - m_i m_i^T)$ obviously is invariant to the sign change of m_i . Geometrically it means that flipping any sensors about the target does not change F .

Let $R \in \mathbb{R}^{d \times d}$ be an orthogonal matrix with $R^T R = I$. Applying R to $\{m_i\}_{i=1}^N$ gives a new FIM as

$$F' = \sum_{i=1}^N c_i^2 [I - R m_i (R m_i)^T] = R F R^T,$$

which is unitarily similar to F . So F and F' have the same eigenvalues, counting multiplicity. Geometrically R represents a rotation ($\det R = 1$), a reflection ($\det R = -1$) or both combined. Thus rotating all sensors the same angle around the target, reflecting all sensors about a line/plane passing through the target, or both combined does not change the eigenvalues of F . ■

The objective function $\|F\|^2$ in (8) is fully determined by the eigenvalues of F . So two placements yield the same $\|F\|^2$ if their FIMs have identical eigenvalues. From Proposition 2, we define the following equivalence relationship.

Definition 1: (Placement Equivalence) Given arbitrary but fixed $\{c_i\}_{i=1}^N$, two placements $\{m_i\}_{i=1}^N$ and $\{m'_i\}_{i=1}^N$ are equivalent if they are only differed by indices permutation, flipping any sensors about the target, or any global rotation, reflection or both combined over all sensors.

Examples of equivalent placements in 2D are shown in Fig. 1.

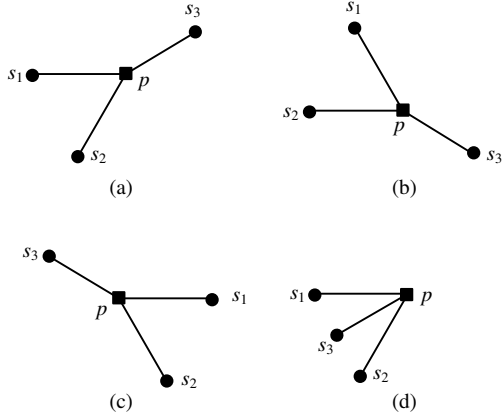


Fig. 1: Examples of equivalent placements ($d = 2, N = 3$): (a) Original placement. (b) Rotate all sensors about the target 60 deg clockwise. (c) Reflect all sensors about the vertical axis. (d) Flipping the sensor s_3 about the target.

IV. OPTIMALITY ANALYSIS

In this section, we show the optimal placements solving (8) consist of two types: regular and irregular. The type is determined by the regularity of the specific sequence $\{c_i\}_{i=1}^N$. Theorem 1 and 2 in this section respectively are generalization of Theorem 8 and 9 in [7].

Theorem 1 (Regular Optimal Placement): In $\mathbb{R}^d$, given a regular sequence $\{c_i\}_{i=1}^N$, a placement $\{m_i\}_{i=1}^N$ is optimal in the sense of solving (8) if and only if

$$\sum_{i=1}^N c_i^2 m_i m_i^T = \frac{1}{d} \sum_{i=1}^N c_i^2 I. \quad (9)$$

Proof: Since $\sum_{i=1}^d \lambda_j = (d-1) \sum_{i=1}^N c_i^2$ is constant, it is straightforward to see

$$\|F\|^2 = \sum_{i=1}^d \lambda_j^2 \geq d \bar{\lambda}^2 = \frac{(d-1)^2}{d} \left(\sum_{i=1}^N c_i^2 \right)^2,$$

where the equality holds if and only if $\lambda_j = \bar{\lambda}$ for all $j \in \{1, \dots, d\}$. Since F is positive definite, $\lambda_j = \bar{\lambda}$ if and only if

$$F = \bar{\lambda} I,$$

from which (9) can be deduced.

Denote $\varphi_i = c_i m_i \in \mathbb{R}^d$, then (9) becomes $\sum_{i=1}^N \varphi_i \varphi_i^T = \frac{1}{d} \sum_{i=1}^N c_i^2 I$ which is the same as (2). From Lemma 1, an optimal placement solving (9) exists if and only if $\{c_i\}_{i=1}^N$ is regular. ■

We call a placement solving (9) as a *regular optimal placement*. Finding a regular optimal placement is equivalent to finding a tight frame. Recall $c_i = 1/(\sigma_i \|r_i\|)$. When $\sigma_i = \sigma_j$ for all $i \neq j$, a regular sequence $\{c_i\}_{i=1}^N$ implies

$$\max_{i=1, \dots, N} \frac{1}{\|r_i\|^2} \leq \frac{1}{d} \sum_{i=1}^N \frac{1}{\|r_i\|^2},$$

which geometrically means no sensor is much closer to the target than the others.

Theorem 2 (Irregular Optimal Placement): In $\mathbb{R}^d$, given an irregular decreasing sequence $\{c_i\}_{i=1}^N$ with irregularity as k_0 , a placement $\{m_i\}_{i=1}^N$ is optimal in the sense of solving (8) if and only if

$$\{m_i\}_{i=1}^N = \{m_i\}_{i=1}^{k_0} \cup \{m_i\}_{i=k_0+1}^N, \quad (10)$$

where $\{m_i\}_{i=1}^{k_0}$ is an orthogonal set with m_i 's mutually orthogonal; $\{m_i\}_{i=k_0+1}^N$ forms a regular optimal placement in the $(d - k_0)$ -dimensional orthogonal complement of $\text{span}(\{m_i\}_{i=1}^{k_0})$.

Proof: Denote $G = \sum_{i=1}^N c_i^2 m_i m_i^T$. Then $F = \sum_{i=1}^N c_i^2 I - G$. Consequently $\|F\|^2 = (d-2)(\sum_{i=1}^N c_i^2)^2 + \|G\|^2$. Hence minimizing $\|F\|^2$ is equivalent to minimizing $\|G\|^2$. Denote $\varphi_i = c_i m_i \in \mathbb{R}^d$. Then $G = \sum_{i=1}^N \varphi_i \varphi_i^T$ with $\|\varphi_i\| = c_i$. Note

$$\begin{aligned} \|G\|^2 &= \text{tr}(G^2) \\ &= \text{tr} \left(\sum_{i=1}^N \varphi_i \varphi_i^T \sum_{j=1}^N \varphi_j \varphi_j^T \right) \\ &= \text{tr} \left(\sum_{i=1}^N \sum_{j=1}^N \varphi_i \varphi_i^T \varphi_j \varphi_j^T \right) \\ &= \sum_{i=1}^N \sum_{j=1}^N (\varphi_i^T \varphi_j)^2. \end{aligned}$$

From (4), $\|G\|^2$ is the frame potential of $\{\varphi_i = c_i m_i\}_{i=1}^N$. From Lemma 4 we have the theorem. ■

We call a placement of the form in (10) as an *irregular optimal placement*. From Theorem 2, finding an irregular optimal placement is equivalent to finding the tightest (but not tight) frame. In $\mathbb{R}^3$ there are at most three orthogonal complements, so only two types of irregular optimal placements exist in 3D space: i) $k_0 = 1$. Then m_1 is orthogonal to $\{m_i\}_{i=2}^N$, and $\{m_i\}_{i=2}^N$ forms a regular optimal placement in a 2D space. ii) $k_0 = 2$. Then m_1 is orthogonal to $\{m_i\}_{i=3}^N$, and m_2 is orthogonal to $\{m_i\}_{i=3}^N$, and $\{m_i\}_{i=3}^N$ are collinear in a 1D space.

Recall $c_i = 1/(\sigma_i \|r_i\|)$. When $\sigma_i = \sigma_j$ for all $i \neq j$, an irregular sequence $\{c_i\}_{i=1}^N$ implies

$$\max_{i=1, \dots, N} \frac{1}{\|r_i\|^2} > \frac{1}{d} \sum_{i=1}^N \frac{1}{\|r_i\|^2},$$

which geometrically means certain sensors are much closer to the target than the others. Then the closest sensors will dominate and drive all the others into an orthogonal complement.

V. EXPLICIT CONSTRUCTION OF REGULAR OPTIMAL PLACEMENTS

From Theorem 2, an irregular optimal placement problem eventually can be converted to a regular one in a lower dimensional space. Therefore, our main concern will be devoted to constructing regular optimal placements.

A. Specific case: $d = 2$ and N is arbitrary

Explicit constructions of 2D optimal placements of two, three and four sensors have been proposed in [7], [8]. But how to construct 2D optimal placements of arbitrary number of sensors with arbitrary sensor-target ranges is not solved yet to our knowledge. We are going to propose a construction method to solve this generic problem. The following lemma can be found in [7], [8], [11], [13], [15].

Lemma 5: In $\mathbb{R}^2$, suppose $m_i = (\cos \theta_i, \sin \theta_i)^T$. Then (9) is equivalent to

$$\sum_{i=1}^N c_i^2 \bar{m}_i = 0, \quad (11)$$

where $\bar{m}_i = (\cos 2\theta_i, \sin 2\theta_i)^T$.

By Lemma 5, the matrix equation (9) is simplified to a vector equation (11). In order to construct an optimal placement in 2D, we first construct $\{\bar{m}_i\}_{i=1}^N$ solving (11) and then retrieve $\{m_i\}_{i=1}^N$ from $\{\bar{m}_i\}_{i=1}^N$. The following theorem gives a necessary and sufficient condition for the existence of solutions to (11). The following theorem is also presented in [7], [8], where sufficiency, however, is not proved. We prove the sufficiency by constructing a solution to (11).

Theorem 3: In $\mathbb{R}^2$, given a sequence $\{c_i\}_{i=1}^N$, there exist $\{\bar{m}_i\}_{i=1}^N$ with $\|\bar{m}_i\| = 1$ solving $\sum_{i=1}^N c_i^2 \bar{m}_i = 0$ if and only if

$$c_j^2 \leq \frac{1}{2} \sum_{i=1}^N c_i^2 \quad (12)$$

for all $j \in \{1, \dots, N\}$.

Proof: Necessity: If $\sum_{i=1}^N c_i^2 \bar{m}_i = 0$, then $c_j^2 \bar{m}_j = -\sum_{i \neq j} c_i^2 \bar{m}_i$ for all $j \in \{1, \dots, N\}$. Hence $c_j^2 = \|\sum_{i \neq j} c_i^2 \bar{m}_i\| = \|\sum_{i \neq j} c_i^2 \bar{m}_i\| \leq \sum_{i \neq j} \|c_i^2 \bar{m}_i\| = \sum_{i \neq j} c_i^2$. Moreover, since $c_j^2 + \sum_{i \neq j} c_i^2 = \sum_{i=1}^N c_i^2$, we have $c_j^2 \leq \frac{1}{2} \sum_{i=1}^N c_i^2$.

Sufficiency: If $c_j^2 \leq \frac{1}{2} \sum_{i=1}^N c_i^2$ for all $j \in \{1, \dots, N\}$, there always exists an index N_0 such that

$$c_1^2 + \dots + c_{N_0-1}^2 \leq \frac{1}{2} \sum_{i=1}^N c_i^2, \quad (13)$$

$$c_1^2 + \dots + c_{N_0-1}^2 + c_{N_0}^2 \geq \frac{1}{2} \sum_{i=1}^N c_i^2. \quad (14)$$

Choose

$$\begin{aligned} l_1 &= c_1^2 + \dots + c_{N_0-1}^2, \\ l_2 &= c_{N_0}^2, \\ l_3 &= c_{N_0+1}^2 + \dots + c_N^2. \end{aligned} \quad (15)$$

Obviously $l_1 + l_2 + l_3 = \sum_{i=1}^N c_i^2$. From (13), $l_1 \leq \frac{1}{2} \sum_{i=1}^N c_i^2$, hence $l_2 + l_3 \geq l_1$. From (14), $l_1 + l_2 \geq \frac{1}{2} \sum_{i=1}^N c_i^2$, hence $l_1 + l_2 \geq l_3$. From $c_{N_0}^2 \leq \frac{1}{2} \sum_{i=1}^N c_i^2$, $l_1 + l_3 \geq l_2$. Therefore, line segments with lengths respectively as l_1, l_2 and l_3 can form a triangle.

Choose $\bar{m}_i = \bar{m}_1$ for all $i \in \{1, \dots, N_0 - 1\}$. Then $\sum_{i=1}^{N_0-1} c_i^2 \bar{m}_i = \sum_{i=1}^{N_0-1} c_i^2 \bar{m}_1 = l_1 \bar{m}_1$. Choose $\bar{m}_i = \bar{m}_N$

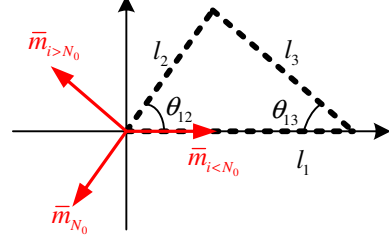


Fig. 2: Geometry of the 2D construction method.

for all $i \in \{N_0 + 1, \dots, N\}$. Then $\sum_{i=N_0+1}^N c_i^2 \bar{m}_i = \sum_{i=N_0+1}^N c_i^2 \bar{m}_N = l_3 \bar{m}_N$. Thus $\sum_{i=1}^N c_i^2 \bar{m}_i = 0$ becomes

$$l_1 \bar{m}_1 + l_2 \bar{m}_{N_0} + l_3 \bar{m}_N = 0.$$

The above equation is solved if and only if $l_1 \bar{m}_1, l_2 \bar{m}_{N_0}$ and $l_3 \bar{m}_N$ form a triangle. To do that, we can choose $\bar{m}_1, \bar{m}_{N_0}$ and $\bar{m}_N$ that align with the three sides of the triangle with side length as l_1, l_2 and l_3 . See Fig. 2. ■

It is notable that (12) actually is a 2D version fundamental inequality, so Theorem 3 is an alternative proof of Lemma 1 in 2D cases.

Once $\{\bar{m}_i\}_{i=1}^N$ solving (11) is obtained, $\{m_i\}_{i=1}^N$ can be retrieved from $\{\bar{m}_i\}_{i=1}^N$. One $\bar{m}_i$ gives two m_i 's with opposite signs, which will lead to equivalent placements. If given an $\bar{m}_i = (\cos 2\theta_i, \sin 2\theta_i)^T$, the corresponding m_i can be chosen as $m_i = (\cos \theta_i, \sin \theta_i)^T$. From the proof of Theorem 3 a construction of 2D regular optimal placements can be summarized as below.

Corollary 1: In $\mathbb{R}^2$, given a regular sequence $\{c_i\}_{i=1}^N$ with $N \geq 3$, an optimal placement can be constructed as below.

- 1) Choose N_0 satisfying (13) and (14). Then compute l_1, l_2 and l_3 in (15).
- 2) Compute internal angles θ_{12} and θ_{13} of the triangle with side lengths as l_1, l_2 and l_3 . See Fig. 2.
- 3) Choose $m_i = (1, 0)^T$ for all $i \in \{1, \dots, N_0 - 1\}$. Choose $m_{N_0} = (\cos \frac{\pi + \theta_{12}}{2}, \sin \frac{\pi + \theta_{12}}{2})^T$ for $i = N_0$. Choose $m_i = (\cos \frac{\pi - \theta_{13}}{2}, \sin \frac{\pi - \theta_{13}}{2})^T$ for all $i \in \{N_0 + 1, \dots, N\}$.

Example 1: In $\mathbb{R}^2$, given 6 sensors with sensor-target ranges as $\|r_1\| = 5, \|r_2\| = 6, \|r_3\| = 7, \|r_4\| = 8, \|r_5\| = 9, \|r_6\| = 10$. The measurement noise variance of each sensor is identical. Then $c_1^2 = 0.0400, c_2^2 = 0.0278, c_3^2 = 0.0204, c_4^2 = 0.0156, c_5^2 = 0.0123, c_6^2 = 0.0100$. We have $\frac{1}{2} \sum_{i=1}^6 c_i^2 = 0.0631$. Because $c_1^2 < \frac{1}{2} \sum_{i=1}^6 c_i^2$ and $c_1^2 + c_2^2 > \frac{1}{2} \sum_{i=1}^6 c_i^2$, choose $N_0 = 2$. Hence $l_1 = 0.0400, l_2 = 0.0278, l_3 = 0.0584$. Then $\theta_{12} = 2.0560$ rad and $\theta_{13} = 0.4344$ rad. As instructed in the construction method, one optimal placement can be computed as $m_1 = (1, 0)^T, m_2 = (0.8563, -0.5165)^T, m_3 = \dots = m_6 = (0.2155, 0.9765)^T$. It is easy to verify that $\sum_{i=1}^6 c_i^2 m_i m_i^T = \frac{1}{2} \sum_{i=1}^6 c_i^2 I$.

B. Distributed Construction

When there are a large number of sensors, it may not be convenient to design the placement for all sensors simultaneously. We shall introduce a distributed construction manner

for large scale sensor placement problems. The following theorem is a generalization of Corollary 7 in [7].

Theorem 4: The union of multiple non-intersecting regular optimal placements in $\mathbb{R}^d$ is still a regular optimal placement in $\mathbb{R}^d$.

Proof: In $\mathbb{R}^d$, suppose there are q regular optimal placements: $\{c_i, m_i\}_{i \in \mathcal{I}_k}$ with $\mathcal{I}_k$ as the index set of the k th placement and $k \in \{1, \dots, q\}$. Different placements have no overlap: $\{c_i, m_i\}_{i \in \mathcal{I}_k} \cap \{c_i, m_i\}_{i \in \mathcal{I}_{k'}} = \emptyset$ for all $k \neq k'$. Define $\mathcal{I} = \bigcup_{k=1}^q \mathcal{I}_k$.

For the k th placement, since $\{c_i, m_i\}_{i \in \mathcal{I}_k}$ is regular optimal in $\mathbb{R}^d$, from Theorem 1 we have

$$\sum_{i \in \mathcal{I}_k} c_i^2 m_i m_i^T = \frac{1}{d} \sum_{i \in \mathcal{I}_k} c_i^2 I.$$

For the union placement, we have

$$\begin{aligned} \sum_{j \in \mathcal{I}} c_j^2 m_j m_j^T &= \sum_{k=1}^q \sum_{i \in \mathcal{I}_k} c_i^2 m_i m_i^T \\ &= \frac{1}{d} \sum_{k=1}^q \sum_{i \in \mathcal{I}_k} c_i^2 I \\ &= \frac{1}{d} \sum_{j \in \mathcal{I}} c_j^2 I. \end{aligned}$$

From Theorem 1, the union placement is optimal in $\mathbb{R}^d$. ■

From Theorem 4, if a large scale regular placement can be divided into some small scale regular subsets, we can design the optimal placement for each subset and then combine them together. Using this idea, a large scale problem can be solved in a distributed manner. So we call this kind of method as *distributed construction*. Because the combination of optimal subsets is arbitrary, there are infinite numbers of optimal placements for the large system. These infinite optimal placements may not be equivalent as in Definition 1, but it easy to show they have the same F .

C. Specific case: Equal-range placements

By equal-range we mean $\|r_1\| = \dots = \|r_N\|$. The corresponding geometry is that all sensors are restricted on the same circle (2D) or sphere (3D) centered at the target. If $\sigma_i = \sigma_j$ for all $i \neq j$, $\|r_1\| = \dots = \|r_N\|$ implies $c_1 = \dots = c_N$. In this case, equal-range placements are all regular. Then (9) can be simplified to $\sum_{i=1}^N m_i m_i^T = \frac{N}{d} I$. Finding an optimal equal-range placement actually is to find a normalized tight frame [11]. We would like to highlight some important properties of equal-range placements in both 2D and 3D.

In $\mathbb{R}^2$, for $N \geq 3$, a placement is optimal if N sensors are located at the vertices of an N -side regular polygon [7], [8], [11], [15]. In addition, it is notable that distributed construction method is very suitable for constructing equal-range optimal placements in both 2D and 3D. In 2D space, any placement with no less than 2 sensors is regular. Any integer larger than 4 can be divided into a linear combination of 2 and 3, so we can always distributedly construct an

optimal placement with $N \geq 4$ using the $N = 2$ and $N = 3$ placements. Consequently, the placements with $N \geq 4$ will always have infinite number of optimal solutions.

In $\mathbb{R}^3$, a placement is optimal if all sensors are located at the vertices of a regular solid [11], [15]. Unfortunately, there are only five regular solids: tetrahedron with four vertices, octahedron with six, cube with eight, icosahedron with twelve and dodecahedron with twenty. But an optimal placement of N sensors can be distributedly constructed by the five regular solids as long as N can be factored into the numbers for which a regular solid exists.

VI. CONCLUSIONS AND FUTURE WORK

This work generalizes the existing work [7], [8] on optimal placement of bearing-only sensors from 2D to 3D. In the future, we may consider to solve the optimal sensor placement problem from a control perspective.

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